

KISAN COMPASS

Farm Decision Intelligence

“Not what to think. What’s actually true — and how sure we are.”

KISAN COMPASS unites real farmer ground truth with crop phenology, Open-Meteo numerical forecasts, AGMARKNET benchmarks, and road logistics to simulate outcomes and guide confident harvest decisions.

TEAM: FusionX4

- **Darshil Nigam — Team Leader**
- Manvi Tripathi
- Ritvika Srivastava
- Devanshu Gupta

THE 5 TRUTH CONTRACTS

- **1. Real Farm Single Source of Truth:** Driven by actual farmer field acreage, crop variety, sowing date, and target yield.
- **2. Deterministic Decision Engine:** Agronomy stage (GDD) and net realization (Price × Qty - Freight) run on pure deterministic math.
- **3. What-If Simulation Sandbox:** Evaluates Sell Now vs Wait vs Sell Part across multiple mandis with uncertainty distributions.
- **4. Transparent Provenance:** Every metric cites source, freshness timestamp, and explicit error bounds ($\pm X\%$).
- **5. Farmer Sovereignty:** The system recommends with explainable rationale; the farmer always decides.

THE PROBLEM: FRAGMENTED DATA VS REAL FARMER DILEMMAS

Fragmented Siloed Information

- **Weather Forecasts:** Generic district-level rain alerts fail to capture farm-level crop vulnerability.
- **Mandi Price Boards:** Raw market rates ignore haulage cost, toll taxes, and transit spoilage.
- **Crop Stage & Soil:** Moisture and ripening status remain trapped in offline observation.
- **Logistics Realities:** Vehicle hiring costs, road conditions, and unloading wait times are completely disconnected.
- **The Trap:** Farmers are overwhelmed with disconnected notifications but zero actionable decision support.

The Farmer's Unanswered Dilemmas

- **Timing Risk:** 'Should I harvest and sell today at ₹2,150/qtl, or wait 5 days for a potential ₹2,320/qtl?'
- **Downside Weather Threat:** 'If 28mm rain arrives in 4 days, will muddy fields trap my tractor and ruin crop moisture?'
- **Mandi Selection:** 'Mandi A pays ₹50 more per quintal than Mandi B, but is 42km farther. Which yields higher NET cash in hand?'
- **Capital Pressure:** 'I have urgent fertilizer bills due this Friday—can I sell 40% now and hold 60% safely?'

The Critical Intelligence Gap

- **No Single Source of Truth:** Existing apps give isolated advisory without knowing the farmer's actual crop cycle or stored stock.
- **Black-Box Hallucinations:** Generic AI chatbots invent fake certainty and dangerous farm advice.
- **Hidden Net Realization:** Farmers suffer distress sales because transport costs and spoilage are never modeled upfront.
- **What Kisan Compass Solves:** Unifies ground truth, context, and economics into a transparent What-If simulation engine.

The Persistent Farm Digital Twin

- **Single Source of Truth:** Persistent profile anchoring real field GPS, acreage (2.5 - 15 Acres), crop variety (Sugarcane HD-2967), sowing date, and available quantity (25 Quintals).
- **Multi-Source Real-World APIs & Models:** Live numerical weather forecasts (Open-Meteo), benchmark mandi prices (AGMARKNET modal rates), and geodesic road freight estimation (1.25x rural detour).
- **Deterministic Agronomy & Economics:** Growing Degree Days (GDD) crop maturity, moisture risk, and true freight costs modeled without hallucination.
- **Human-in-the-Loop Sovereign Control:** 'The system recommends with transparent evidence. The farmer makes the final decision.'
- **Closed-Loop Evolution:** Every decision is recorded into decision memory to calibrate future advice against actual harvest outcomes.

10 KEY PLATFORM FEATURES (CAPABILITY MATRIX)

- **1. FARM DIGITAL STATE:** Persistent database storing farmer profile, field boundaries, soil class, and verified acreage.
- **2. CROP AGRONOMY:** Thermal time / GDD phenology tracking growth stages from vegetative to harvest readiness.
- **3. WEATHER INTELLIGENCE:** 7-day numerical forecasts with rain probability, wind speed, and soil moisture trends.
- **4. MARKET INTELLIGENCE:** Official AGMARKNET mandi modal prices, arrival volumes, and 30-day historical trends.
- **5. NET REALIZATION:** Real take-home profit computed dynamically (Modal Rate × Qty - Road Freight - Mandi Cess).
- **6. WHAT-IF SIMULATION:** Sandbox evaluating Sell Now vs Wait vs Sell Part across multiple mandis with risk bounds.
- **7. DECISION EXPLANATION:** AI reasoning strictly confined to explaining verified numbers with cited evidence.
- **8. CONTINUOUS FARM WATCH:** Background monitor evaluating weather shifts and price surges to trigger threshold alerts.
- **9. EXECUTION INTELLIGENCE:** Action checklist (labor booking, vehicle hiring, mandi pass, transit moisture protection).
- **10. DECISION MEMORY:** Audit trail tracking farmer actions, actual realized prices, and model accuracy calibration.

TECHNICAL APPROACH: SYSTEM ARCHITECTURE & ENGINE SEPARATION

END-TO-END DATA FLOW PIPELINE

FARMER DATA → PERSISTENT FARM STATE → CONTEXT ENGINES → DETERMINISTIC ENGINE → WHAT-IF SIMULATION → EVIDENCE & AI → FARMER DECISION → EXECUTION & OUTCOME

DETERMINISTIC ENGINE (Pure Math & Verification)

- **Role & Scope:** Handles 100% of calculations, agronomy GDD, economics, logistics formulas, and simulation outcome bounds.
- **Technology:** TypeScript & Python calculation core running verifiable mathematical routines.
- **Input Sources:** Real farmer field data (acreage, crop variety) + Open-Meteo numerical weather + AGMARKNET mandi modal prices.
- **Economic Logic:** Computes Net Cash = (Modal Price × Qty) - (Freight per km × Distance + Unloading Fee).
- **Zero Hallucination Guarantee:** No generative AI can tamper with or fabricate financial numbers or crop stages.

AI / LLM EXPLANATION LAYER (Strictly Constrained)

- **Role & Scope:** Natural-language explanation, farmer conversational interface, and multilingual regional translation.
- **Strict Evidence Binding:** Reads structured JSON outputs from the deterministic engine; strictly forbidden from calculating numbers.
- **Verifiable Rationale:** Translates trade-offs into plain farmer dialect: 'Wait recommendation is driven by +8% price upside vs 28mm rain on Day 4.'
- **Transparent Uncertainty:** Always cites data source, freshness timestamp, and confidence error bounds (±X%).
- **Human-in-the-Loop:** Empowers farmer judgment with clear reasoning—never dictates autonomous actions.

TECH STACK — REAL-WORLD PRODUCTION ARCHITECTURE

React 18 + TypeScript • Supabase / PostgreSQL (Row-Level Security) • Open-Meteo NWP Forecast API • AGMARKNET / DMI Mandi API • Deterministic Agronomy & Economic Engine • Grounded LLM Explanation Layer

WHAT-IF SIMULATION: EVALUATING SCENARIO TRADE-OFFS

SCENARIO A: SELL NOW

- **Action:** Immediate harvest and dispatch to closest regional Mandi.
- **Gross Rate:** ₹2,150 / Quintal (Current modal benchmark).
- **Transport Cost:** ₹1,800 (18km diesel + vehicle hire).
- **Net Realization:** ₹51,950 for 25 Quintals.
- **Risk Exposure:** ZERO weather exposure; immediate working capital.
- **Uncertainty:** $\pm 2.5\%$ (Very tight confidence bound).

SCENARIO B: WAIT 5-7 DAYS

- **Action:** Delay harvest to capture anticipated festival price rebound.
- **Projected Rate:** ₹2,320 / Quintal (+7.9% market upside).
- **Weather Risk:** 65% probability of 28mm precipitation on Day 4.
- **Downside Penalty:** Field waterlogging could delay cutting by 4 days, causing ₹3,500 moisture dockage.
- **Expected Net:** ₹54,500 (Range: ₹48,000 – ₹58,000).
- **Uncertainty:** $\pm 11.8\%$ (Wide risk variance).

SCENARIO C: SELL PARTIALLY

- **Action:** Staggered hedge — liquidate 60% immediately, hold 40%.
- **Immediate Cashflow:** ₹31,170 secured for immediate input loan repayment.
- **Upside Participation:** 40% volume retains exposure to potential price surge.
- **Downside Buffer:** Rain damage risk halved on held stock.
- **Expected Net:** ₹53,400 with bounded worst-case floor.
- **Uncertainty:** $\pm 5.2\%$ (Recommended for balanced risk profile).

“We don't pretend to know exactly what the future holds. We compute how the outcome distribution changes under each decision so the farmer can choose with open eyes.”

TRUST, EXPLAINABILITY & GRACEFUL FAILURE HANDLING

Truthful Terminology & Honest Labeling

- **Numerical Weather Forecast:** Clearly labeled 'Open-Meteo 7-day numerical forecast' — NEVER mislabeled as 'live satellite radar'.
- **Official Market Benchmarks:** Labeled 'AGMARKNET / DMI benchmark price' — NEVER fabricated as a 'live Wall Street stock ticker'.
- **Calculated Net Realization:** Defined as '(Modal Rate × Quantity) - (Freight + Unloading)' — NEVER marketed as 'guaranteed profits'.
- **Estimated Road Routing:** Calculated via geodesic Haversine distance with 1.25x rural detour factor — NEVER faked as 'hardware GPS tracker telemetry'.
- **Agronomy Phenology:** Derived from Growing Degree Day (GDD) base temperatures — NEVER claimed to be 'mind-reading AI'.

The Five Strict Truth Contracts

- **1. Provenance Integrity:** Every metric exposes its data source, mathematical formula, and fetch timestamp.
- **2. Transparent Uncertainty:** Every financial prediction displays an explicit confidence bound ($\pm X\%$) instead of false certainty.
- **3. Explainable Rationale:** AI recommendations cite verified numbers (e.g. 'Rain probability 65% on Oct 2 exceeds 40% threshold').
- **4. Human Sovereignty:** Recommendations are advisory; the farmer retains unconstrained freedom to execute or override.
- **5. Calibration Memory:** Realized harvest results are audited against predictions to continuously self-calibrate.

WHEN DATA FAILS: 3-TIER GRACEFUL FAILURE PROTOCOL

- API AVAILABLE → Live fresh data fetched + highest confidence score ($\pm 2.5\%$).
- API UNAVAILABLE → Fallback to cached / last-known values + prominent freshness warning + reduced confidence score ($\pm 15\%$).
- NO RELIABLE DATA → Explicit fallback prompt requesting farmer input — ZERO hallucinated or fabricated numbers.

“CORE OPERATIONAL MANDATE: Failure lowers confidence — never truthfulness.”

Zero-Hardware Feasibility & Scalability

- **Zero Field Hardware Needed:** Requires no expensive IoT soil probes or drone subscriptions; runs as a lightweight responsive PWA on any smartphone.
- **Ground Truth Driven:** Automatically anchors to the farmer's self-entered field size, crop variety, sowing date, and local village.
- **Production Cloud Architecture:** Built with Vite/React frontend and Supabase PostgreSQL with strict Row-Level Security (RLS) guaranteeing total farmer privacy.
- **Low-Bandwidth Optimization:** Compact JSON payloads (<45KB) ensure ultra-fast load times even on spotty 2G/3G rural networks.
- **Field Pilot Readiness:** Fully tested with real farmer data (Darshil Nigam, Kanpur Nagar, Sugarcane HD-2967, 25 Quintals).

Measurable Farmer Transformation

- **Information Parity:** Transforms fragmented apps into a single coherent decision cockpit.
- **Protected Net Income:** Eliminates distress sales by accounting for freight costs and transport timing upfront (+12% to +18% net realization).
- **Mitigated Weather Loss:** Saves entire harvests from unseasonal rainfall through proactive 72-hour What-If alerts.
- **Transparent Trust:** Replaces opaque black-box AI with honest uncertainty ranges and verifiable government data sources.
- **Farmer Self-Reliance:** Equips smallholders with institutional-grade decision intelligence.

PERSISTENT FARM STATE

- Maintains farmer profile, field boundaries, crop stage, and available quantity as one evolving context.
- Never asks for repeated data; grows with the crop cycle.

WHAT-IF SIMULATION

- Simulates Sell Now vs Wait vs Split with explicit market upside vs rain downside trade-offs.
- Quantifies uncertainty bounds rather than false certainty.

EVIDENCE-FIRST AI

- Every number is connected to API origin, mathematical formula, freshness age, and confidence.
- Zero black-box hallucinations; citable explanations.

HUMAN-IN-THE-LOOP

- The system calculates and recommends with transparent evidence.
- The farmer retains 100% executive authority to decide and execute.

Traditional Fragmented Approach (The Silo Problem)

- **Fragmented Point Solutions:** Weather app + Mandi SMS + WhatsApp PDF advisory completely disconnected.
- **Stateless & Forgetful:** Each interaction starts from zero; no memory of field conditions or previous cuts.
- **Hidden Freight Costs:** Unanticipated transport and toll costs trigger distress sales at the mandi gate.
- **Unchecked Hallucinations:** Generic AI chatbots invent fake certainty and dangerous agronomy advice.

KISAN COMPASS: Continuous Closed-Loop Intelligence

- 1. Ground Truth → Persistent farm profile & real field boundaries.
- 2. Context Aggregation → Live weather models, mandi benchmarks & road routing.
- 3. Deterministic Simulation → Verified mathematical optimization & risk bounds.
- 4. Human-in-the-Loop → Explainable advisory; farmer retains complete decision authority.
- 5. Action Execution → Logistics checklist (labor, vehicle, mandi token, moisture test).
- 6. Outcome Calibration → Realized harvest prices logged to refine future predictions.

Our Product Manifesto

- “Not what to think. What’s actually true — and how sure we are.”
- **Empowering Smallholder Farmers:** Indian agriculture doesn't need another generic weather widget or chat bot. It needs decision clarity.
- **Ground Truth as Anchor:** Every calculation begins with what the farmer actually planted and what is currently in the soil.
- **Truth Over Hype:** We never disguise forecasts as certainty, never hallucinate live tickers, and always show confidence bounds.
- **Farmer in the Driver's Seat:** Technology should illuminate trade-offs, not dictate commands.
- **Tested, Deployed & Audited:** Connected to real Supabase database with production RLS security and verified deterministic calculations.

TEAM FUSIONX4

- **Team Leader:** Darshil Nigam
- **Core Member:** Manvi Tripathi
- **Core Member:** Ritvika Srivastava
- **Core Member:** Devanshu Gupta
- **Project Repository:** KISAN COMPASS (Farm Decision Intelligence)
- **Architecture:** React 18 • TypeScript • Supabase RLS • Open-Meteo • AGMARKNET • Geodesic Haversine Routing
- **Ready for Demonstration:** Live database, live decision engine, and complete judge walkthrough.